A CONWAY 99-GRAPH, IF IT EXISTS, IS ASYMMETRIC: STATEMENT AND DEPENDENCIES

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Abstract. A Conway 99-graph is a strongly regular graph with parameters $(99, 14, 1, 2)$; whether one exists is open. Assembling published results with three recent manuscripts, which exclude automorphisms of order 2, 3 and 7, one obtains that such a graph has no nontrivial automorphism. This note states that consequence and lists, prime by prime, the result it uses, the nature of its proof (by hand, by computer, formalised or not) and its status. It contains no new mathematics. The statement is conditional on these ingredients as they stand; in particular the manuscript on order 2 has one validation of its computations still to be completed.

1. Statement

Let Γ be a strongly regular graph with parameters $(99, 14, 1, 2)$ and $G = \mathrm{Aut}(\Gamma)$.

Theorem 1. *Assume the results* (A)–(E) *of Section 2. Then G is trivial: a strongly regular graph with parameters* $(99, 14, 1, 2)$*, if it exists, is asymmetric.*

Proof. If $G \neq 1$, it has an element of prime order p (Cauchy). By (A), $p \in \{2, 3, 7, 11\}$. These four primes are excluded by (B), (C), (D) and (E) respectively. □

2. The ingredients

(A) $|G|$ *divides* $2 \cdot 3^3 \cdot 7 \cdot 11$. Makhnev and Minakova [MM04], by character theory. Published; no computer.

(B) *No automorphism of order* 2. The paper [C2]. The first half (an involution gives a frame of 51 vectors in one of the 156 odd unimodular lattices of rank 24 without vectors of norm 1) is proved by hand. The second half (no such frame in any of the 156 lattices) is computer-assisted, and verified as described in Section 9 of [C2]; in the present draft of [C2] one validation, the recertification of a part of the transfers between lattices, remains to be completed. Not refereed; not formalised.

(C) *No automorphism of order* 3. The manuscript [D3]: linear algebra over $\mathbb{Z}$ and the Eisenstein integers, without computer. An automorphism of order 3 has 0 or 3 fixed points (a count that reproves a result of [MM04]); in both cases two lattices of eigenvectors must have determinants congruent modulo 3, and they are not. Formalised in Lean 4 with Mathlib (Appendix A); the formalisation was read, not rebuilt, for this note. Not refereed. Independently, the case of an automorphism of order 3 with fixed points is excluded by computer in [B09, BL11].

(D) *No automorphism of order* 7. Behbahani and Lam [B09, BL11], by a computer enumeration of orbit matrices; published. Without a computer the case was open: Cesarz and Woldar [CW25] prove that $G \cong \mathbb{Z}_7$ if 7 divides $|G|$ (published), which does not exclude an automorphism of order 7. The manuscript [D7] excludes it by hand, starting from

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the orbit structure and the counting identities of [CW25]; it is formalised in Lean 4 with Mathlib (Appendix A; read, not rebuilt, for this note). Not refereed.

(E) *No automorphism of order* 11. Behbahani and Lam [B09, BL11], by computer; published. A report of the MIT PRIMES programme [LM14] gives an elementary argument, which was not verified for this note.

prime	excluded by	method	formalised	status
$p \neq 2, 3, 7, 11$	[MM04]	characters, by hand	no	published
2	[C2]	by hand, then computer-assisted	no	manuscript
3	[D3]	by hand	Lean 4	manuscript
7	[BL11]	computer	no	published
	[D7], using [CW25]	by hand	Lean 4	manuscript
11	[BL11]	computer	no	published

TABLE 1. What Theorem 1 uses.

3. Remarks

Remark 2 (What is known without any computer). From [MM04], [CW25], [D3] and [D7]: $|G|$ divides $2 \cdot 11$, and if $|G|$ is even it divides 6 [CW25]; hence $|G| \in \{1, 2, 11\}$. The two remaining values are excluded by the computations of [BL11] (order 11) and of [C2] (order 2).

Remark 3 (Before these manuscripts). The published results [MM04, Mak10, B09, BL11, CM20, CW25] leave $|G| \in \{1, 2, 3\}$, the open cases being an involution and a fixed-point-free automorphism of order 3. The manuscript [D3] removes 3 and [C2] removes 2; the results of [CM20] on the orders 6 and 9 are then no longer needed.

Remark 4 (Formalisation). Theorem 1 itself is not formalised: (A), (B) and (E) are admitted. A formal statement "the automorphism group is trivial" would need (B) at least as a named hypothesis.

Remark 5 (Consequence for searches). Every search for a Conway 99-graph that prescribes a nontrivial automorphism is bound to fail: a proof of existence or of nonexistence has to deal with a graph without symmetry.

Appendix A. The formal statements

Order 3. Formalised in Lean 4 with Mathlib: the statement file `Challenge.lean` imports only Mathlib and contains

```
theorem srg99_no_orderOf_three {V : Type*} [Fintype V] (G : SimpleGraph V)
[DecidableRel G.Adj] (hG : G.IsSRGWith 99 14 1 2) (σ :  G ≃g G) :
orderOf σ ≠ 3
```

(namespace `Conway99`; Lean `v4.35.0-rc2`). According to the repository, the proofs use no `sorry` and only the axioms `propext`, `Classical.choice`, `Quot.sound`, and `lake comparator` checks that the proved statements are those of the statement file; its verification workflow (clean rebuild, comparator, and five further checkers) reports success on 23 September 2026, at commit `874903ff`, and the three later commits change no Lean file. For this note the statement file and the description were read; the formalisation was not rebuilt.

Order 7. [D7] is formalised in Lean 4 with Mathlib (`theorem srg99_no_orderOf_seven`, same form as above with 7; no `sorry`, the same three axioms, `lake comparator`; verification workflow successful on 2 October 2026 at commit `8c6ce690`, no Lean file changed since; read, not rebuilt, for this note).

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References

[B09] M. Behbahani, *On strongly regular graphs*, Ph.D. thesis, Concordia University, Montréal, 2009.

[BL11] M. Behbahani and C. Lam, *Strongly regular graphs with non-trivial automorphisms*, Discrete Math. **311** (2011), no. 2–3, 132–144.

[C2] D. Djematene, *A strongly regular graph with parameters* $(99, 14, 1, 2)$ *has no automorphism of order two*, manuscript and supplement, October 2026, `https://github.com/amazing-source/strongly-regular-graphs-research/tree/main/papers/conway99_order2`; programs, records and certificates available from the author on request.

[CM20] D. Crnković and M. Maksimović, *Construction of strongly regular graphs having an automorphism group of composite order*, Contrib. Discrete Math. **15** (2020), no. 1, 22–41.

[CW25] P. G. Cesarz and A. J. Woldar, *On the automorphism group of a putative Conway 99-graph*, Algebr. Comb. **8** (2025), no. 2, 379–398.

[D3] D. Djematene, *Le graphe de Conway* $\mathrm{srg}(99, 14, 1, 2)$ *n'a pas d'automorphisme d'ordre* 3*: une obstruction par les discriminants*, manuscript with a formalisation in Lean 4, September 2026, `https://github.com/amazing-source/strongly-regular-graphs-research/tree/main/papers/conway99_order3`; Lean formalisation available from the author on request.

[D7] D. Djematene, *A computer-free proof that a Conway* 99*-graph has no automorphism of order seven*, manuscript with a formalisation in Lean 4, October 2026, `https://github.com/amazing-source/strongly-regular-graphs-research/tree/main/papers/conway99_order7`; Lean formalisation available from the author on request.

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